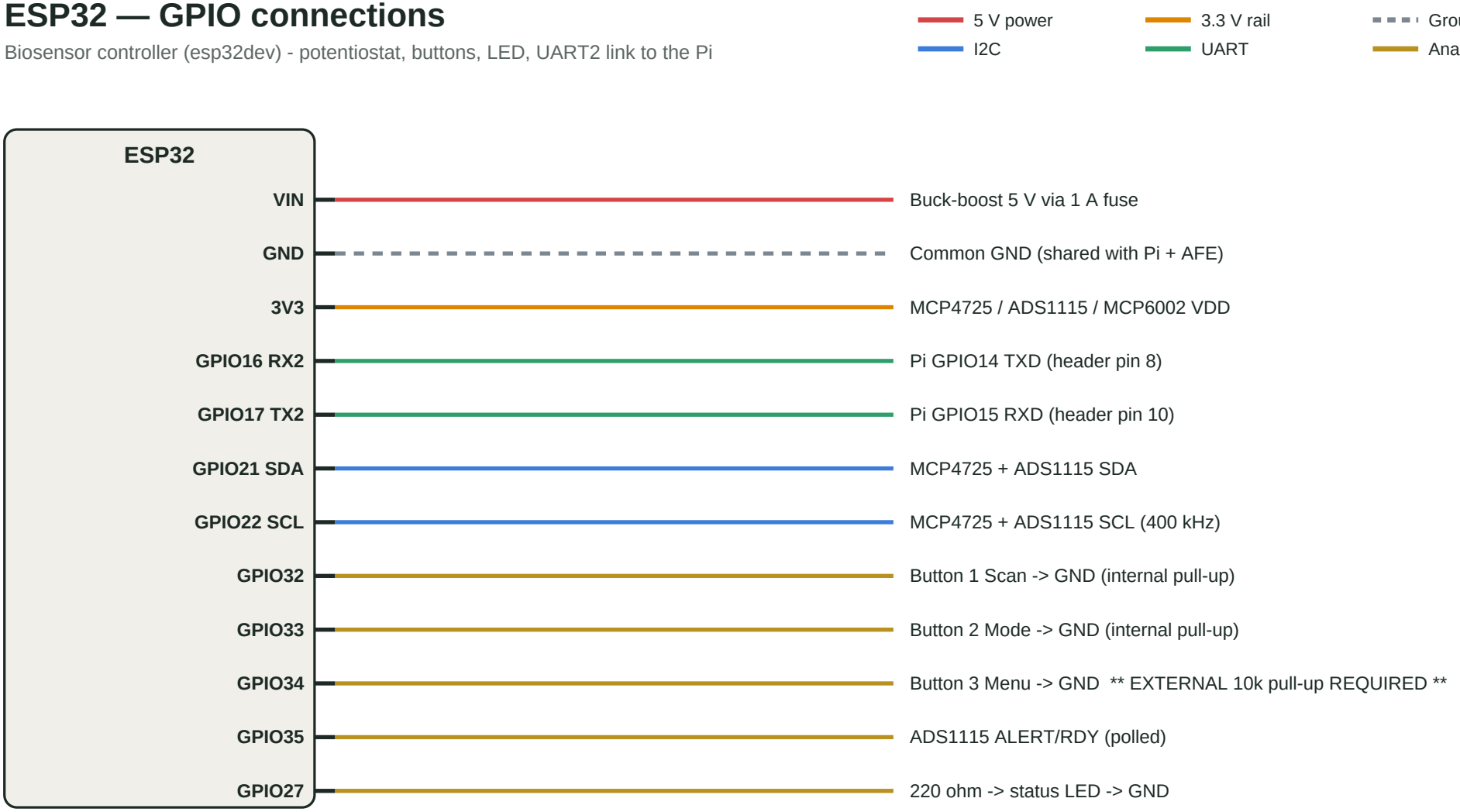
ESP32 — GPIO connections

Biosensor controller (esp32dev) - potentiostat, buttons, LED, UART2 link to the Pi



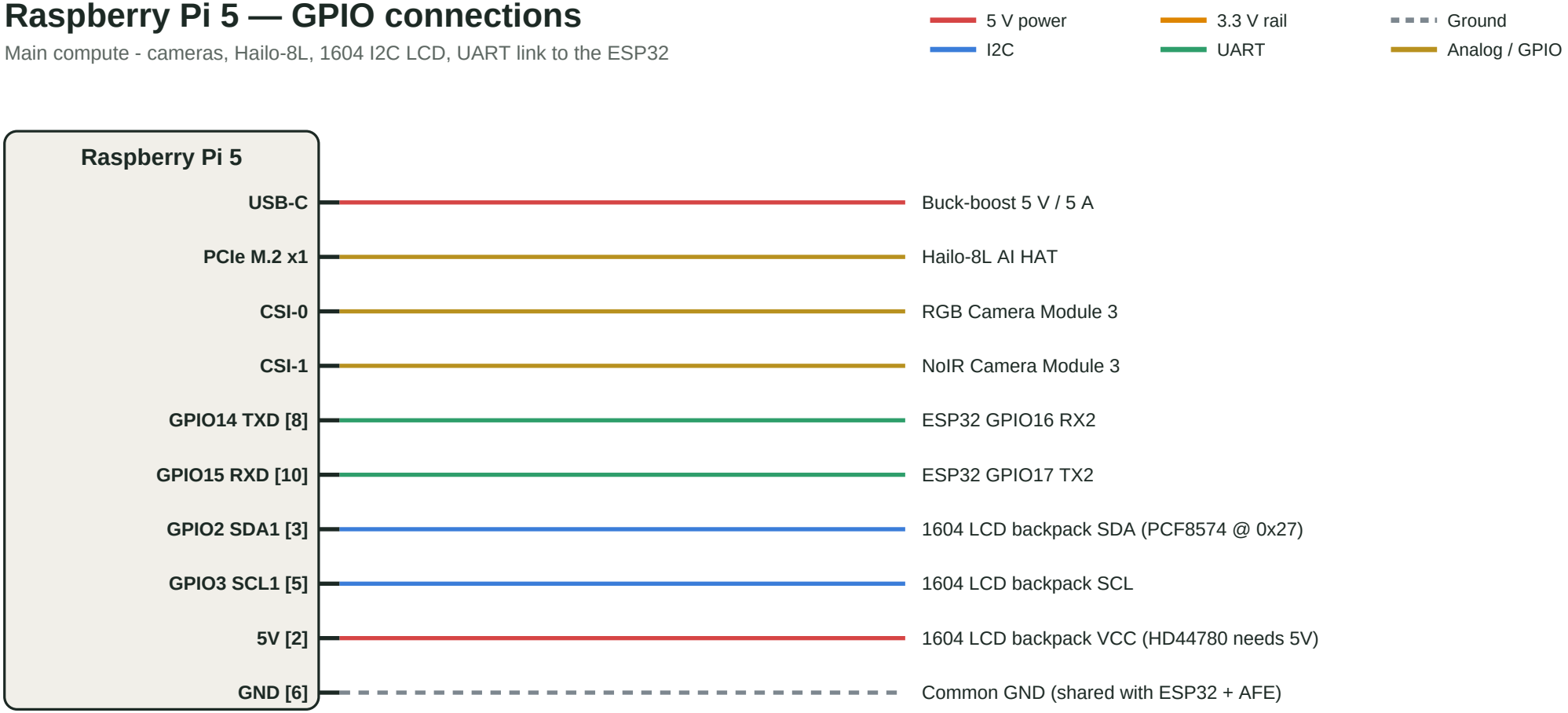
Notes

- GPIO34 and GPIO35 are input-only and have no internal pull-ups.
- I2C addresses: MCP4725 = 0x60 (ADDR->GND), ADS1115 = 0x49 (ADDR->3.3V).
- UART2 crosses over: TX->RX both directions. 115200 8N1, text protocol.

Sources: esp32/src/config.h, pi5/config.py, docs/WIRING.md - regenerate with docs/generate_gpio_pinout.py. Common ground across all boards is mandatory.

Raspberry Pi 5 — GPIO connections

Main compute - cameras, Hailo-8L, 1604 I2C LCD, UART link to the ESP32



Notes

- [n] = physical 40-pin header position. BCM numbering elsewhere.
- Remove the LCD backpack's 5V SDA/SCL pull-ups (or level-shift): Pi I2C pins are 3.3V; the Pi's onboard 1.8k pull-ups suffice.
- GPIO8/10/11/18/24/25 freed by the ILI9341 -> 1604 LCD change.
- UART link is 3.3V logic on both ends - no level shifter needed.

Sources: esp32/src/config.h, pi5/config.py, docs/WIRING.md - regenerate with docs/generate_gpio_pinout.py. Common ground across all boards is mandatory.